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How Behavioral Exposure Reveals Privacy Risks Beyond Location Uniqueness

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Abstract. Individual-level mobility data enables valuable applications but raises significant privacy risks due to the distinctive nature of human mobility and susceptibility to identification. This creates a fundamental challenge: quantifying user exposure, i.e., the degree to which individuals can be distinguished, while supporting both privacy protection and data utility. Existing work focuses on location-based and spatio-temporal exposure, emphasizing visited places and trajectory distinctiveness. In this paper, we propose a complementary perspective based on persistent mobility behaviors, such as routines and mobility preferences, which can make users distinctive beyond raw locations. We introduce `HEXPOSE` (`Hyperbox-based Exposure in a Mobility Behavioral Space`), a framework that represents users in a multidimensional behavioral space and defines individualized indistinguishability regions to quantify exposure. Our approach provides an interpretable, data-driven characterization of privacy risk without relying on specific, predefined adversarial models. Through empirical analysis of two large-scale real-world cellular datasets, we show that behavioral exposure reveals vulnerable users and patterns that are not captured by state-of-the-art location-based approaches. Our findings highlight behavioral exposure as a complementary dimension of privacy risk in mobility data.

1 Introduction

Mobility has always been a defining trait of the human species [18]. In the modern era, it has become digitally traceable: individual-level mobility data supports advanced services and research [8, 16] and generates billions in revenue (projected US $462B by 2032) [2]. At the same time, mobility traces are highly distinctive and pose serious privacy risks. These risks are reinforced by persistent routines and time-based social practices (e.g., work and family time) that remain stable in everyday life [17]. Prior work shows that a small number of spatiotemporal points can uniquely identify over 90% of users in large urban settings [11, 13, 32], highlighting the inherent vulnerability of mobility datasets.

Traditional privacy-preserving techniques can mitigate identification risks, but they often enforce uniform protection mechanisms that ignore individual

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differences and can unnecessarily degrade utility [16, 20]. In practice, privacy risk varies across users and is shaped by mobility patterns [11, 14]. This motivates flexible methods that measure and interpret such risk at the individual level, enabling targeted protection while preserving mobility-driven services.

In this work, we refer to this individual-level privacy risk as *user exposure*, that is *the degree to which an individual can be distinguished from others in a dataset based on their mobility patterns*. A highly exposed user exhibits distinctive mobility characteristics that make them stand out from the population, increasing their susceptibility to identification, while a less exposed user is harder to identify due to more common mobility patterns. Re-identification attacks build on this notion by combining identification with auxiliary information to link users to external identities. In this sense, *exposure* captures the underlying distinctiveness that such attacks may exploit and provides a relative measure of privacy risk to guide targeted protection strategies.

From this perspective, existing approaches capture different facets of user exposure. A prominent line of work focuses on *location-based exposure*, which relies on visited locations and spatiotemporal trajectories. A representative example is *Uniqueness* [11], which measures how much an individual's ordered spatiotemporal trace stands out from others. Other works estimate exposure through the success of predefined location-based re-identification attacks, showing that mobility metrics (e.g., entropy, radius of gyration) can help predict this attack success [23, 26, 28], but remain tied to specific adversarial settings.

We instead consider a complementary perspective that shifts the focus from *where* and *when* users visit locations to *how* they move, capturing persistent behavioral traits rather than raw trajectories. Precisely, we define exposure directly from users' mobility behavior, referred to as *behavioral exposure*, i.e., how much a user stands out based on mobility characteristics. For instance, a user may visit common locations shared by many others, yet exhibit distinctive mobility patterns (e.g., a tourist among regular commuters).

In this direction, we previously introduced MoBES [14], which quantifies *behavioral exposure* in a multidimensional space, i.e., a space that considers several mobility metrics, without assuming a specific adversarial model. However, MoBES relies on an aggregated score that can overlook users who stand out in only a few mobility characteristics, as these distinctions may be diluted when combined with more common behaviors. *This motivates the need for an approach that can capture such individual dimensions of distinctiveness and provide a more precise and interpretable assessment of exposure.*

This paper introduces `HEXPOSE` (*Hyperbox-based Exposure in a Mobility Behavioral Space*), a new framework for measuring *behavioral exposure* in mobility data. Like prior unsupervised works, `HEXPOSE` does not assume a specific adversarial strategy. However, it provides a more structured and fine-grained characterization of exposure, capturing distinct behaviors along specific mobility dimensions that aggregation-based methods may miss. This leads to a more accurate assessment of user risk and enables understanding *why* users are exposed for targeted privacy protection. Overall, we make three main contributions:

Conceptual: We formalize *behavioral exposure* as a privacy risk complementary to location-based exposure. While the latter relies on the uniqueness of visited locations or trajectories, behavioral exposure captures how distinctive a user's mobility characteristics are through a set of mobility metrics (e.g., regularity, diversity, spatial range, cf. §3). `HEXPOSE` quantifies this distinctiveness at the individual level and attributes it to the behavioral factors that contribute to it.
Methodological: We introduce `HEXPOSE` as a general, multidimensional, and interpretable behavioral exposure framework. Each user is represented in a behavioral space defined by mobility metrics, and an individualized *hyperbox* is constructed using per-dimension tolerance thresholds. This hyperbox defines a similarity region: users with many neighbors inside it are less exposed, while users with few or no neighbors are more exposed. Unlike aggregation-based approaches such as MoBES, `HEXPOSE` quantifies exposure in a way that remains sensitive to localized deviations across individual dimensions and provides interpretable insights into which behavioral factors contribute to exposure (cf. §3).
Empirical: We conduct a large-scale empirical study of `HEXPOSE` on two anonymized Call Detail Record (CDR) datasets from the cities of Shanghai and Shenzhen, in China (cf. §4). Our results show that behavioral exposure is uneven across users and depends on similarity thresholds and dataset characteristics. `HEXPOSE` identifies exposed users overlooked by MoBES and reveals exposure patterns not captured by location-based methods such as Uniqueness. We further relate behavioral exposure to a simple re-identification setting, showing that it captures complementary vulnerabilities not directly exploited by existing attacks (cf. §5).

Overall, these contributions make `HEXPOSE` a diagnostic tool for data owners, allowing them to identify which users are behaviorally exposed and why, and to inform targeted privacy protection strategies before data publishing.

The rest of this paper is organized as follows: §2 reviews privacy risks in mobility data and related exposure measures. §3 introduces behavioral exposure and the `HEXPOSE` framework. §4 presents the experimental setup, and §5 reports empirical findings. §6 concludes the paper and discusses perspectives.
Reproducibility: To support reproducibility, we release the full implementation of `HEXPOSE` along with all evaluation scripts used in this paper at [1].

2 The landscape of privacy risks in mobility data

The increasing availability of mobility data has enabled companies to design and evaluate data-driven services, resulting in a mobility-based ecosystem valued at billions of dollars [2]. This ecosystem could support a broader range of services and grow in value even further if such data were more accessible. However, due to the inherent sensitivity of mobility data, it is not easily shared. Indeed, several works in the literature have shown that only a few spatiotemporal points are sufficient to uniquely identify large portions of users in a mobility dataset [11, 13, 32]. Such discouraging results have led to a growing body of research on

anonymizing such data, using techniques such as k-anonymity [16] or Differential Privacy [20]. Anonymization approaches, however, face two main limitations: (*i*) they often cause substantial utility loss, especially when applied uniformly [20], and (*ii*) they are computationally expensive, limiting scalability [7]. Aggregated data offers an alternative but reduces flexibility across applications.

Trends in exposure quantification: To address these challenges, prior work has proposed methods [11, 14, 23, 26] to quantify *user exposure*, i.e., the extent to which individuals can be distinguished based on their mobility patterns. These approaches can be broadly grouped into methods capturing (i) *location-based exposure*, operating on raw trajectories, and (ii) *behavioral exposure*, operating on behavior representations (i.e., mobility metrics) derived from raw trajectories.

The most prominent method for location-based exposure, Uniqueness [11], measures exposure by identifying sub-trajectories, i.e., ordered location sequences within a time window t_w, that are unique across users. Users with non-repeated sub-trajectory patterns are considered exposed.

In contrast, several works leverage mobility metrics (e.g., entropy, regularity, radius of gyration, stationarity, diversity) [9, 15, 22, 29] within specific adversarial settings. These approaches implement location-based re-identification attacks and measure their success at the individual level, showing that mobility metrics can help predict this success [23, 26]. While effective, they remain tied to predefined attacks and do not directly define exposure as a standalone notion. Nonetheless, they highlight that mobility behavior carries information relevant to users' distinguishability beyond raw trajectories.

More recently, MoBES [14] develops this intuition into a behavioral exposure framework. It represents users in a multidimensional space defined by mobility metrics and quantifies a user's exposure as the distance to her nearest neighbors. In this formulation, exposure is computed as an aggregated measure across metrics, which can mask strong distinctions arising from a small number of dimensions, as these may be diluted when combined with more common dimensions. As a result, exposure driven by a few distinctive traits may be overlooked.

Positioning: We propose `HEXPOSE`, a novel framework for quantifying *behavioral exposure* that supports generalizable risk assessment without relying on predefined adversarial models. Unlike aggregation-based approaches such as MoBES, `HEXPOSE` captures distinctions that arise from a few mobility characteristics, preventing them from being diluted by more common behaviors. This results in a more precise and interpretable assessment of user exposure.

3 Shaping Behavioral Exposure

We describe how `HEXPOSE` models user behavior and quantifies exposure. The approach represents each user in a multidimensional space defined by mobility metrics and constructs a personalized hyperbox that defines a similarity region. Exposure is then assessed based on the number of behaviorally similar neighbors within this region: fewer neighbors indicate higher exposure.

3.1 Foundations and design

We consider a dataset $D = \big(U, \Phi\big)$, where $U = \{1, 2, \ldots, |U|\}$ is the set of users and Φ a set of user trajectories. Each user $i \in U$ has one trajectory $\Phi_i \in \Phi$. Each trajectory Φ_i is composed of tuples (i, t, l), where i refers to the user, t to the timestamp of the record, and l to the location (e.g., longitude and latitude) of the user i at time t.

To analyze user mobility, we extract from each trajectory Φ_i a set of mobility metrics, $M = \{m_1, m_2, \ldots, m_{|M|}\}$, where m_r denotes the r-th mobility metric. These metrics capture various individual characteristics of mobility behavior (e.g., number of unique locations and entropy). Each user i is then represented in a $|M|$-dimensional behavioral space by a vector: $(m_1^i, m_2^i, \ldots, m_{|M|}^i) \in \mathbb{R}^{|M|}$, where m_r^i is the value of metric m_r derived from trajectory Φ_i. This representation enables a compact and structured model to represent user mobility behavior. The primary objective of this representation is to find users whose mobility patterns deviate substantially from the majority of behaviors. Such users tend to be farther from their neighbors in one or more dimensions of the behavioral space and are therefore more likely to be uniquely identified.

To assess the exposure of a user i, we define a *personalized similarity region in the multidimensional behavioral space*, centered at the **user i's position** $(m_1^i, m_2^i, \ldots, m_{|M|}^i)$. This region is modeled as a **hyperbox** (i.e., a rectangular parallelepiped), where the extent along each dimension r is governed by a **tunable parameter** $v \in \mathbb{R}_+$, representing the allowable **relative** (i.e., percentwise) deviation or tolerance per metric. Formally, for each dimension r, the bounds of the hyperbox are given by $(1 \pm v) \times m_r^i$. A user $j \in U$ is considered a *neighbor* of user i if j's metric vector representation lies within i's hyperbox.

Note that, by adopting a percentwise deviation with respect to i's behavior, as captured by each metric m_r^i, the interpretation of parameter v does not depend on any specific metric or value ranges. It should be interpreted as *the maximum allowable relative deviation (tolerance) in each behavioral metric (dimension) of i, for j to be considered similar (i.e., indistinguishable) to i.* Thus, the value of v *quantifies the strictness of similarity*: smaller values imply tighter bounds and more stringent criteria for behavioral similarity. While different values of v could be adopted for each dimension (e.g., v_r for metric m_r), we choose here to adopt the same value v to facilitate interpretability of the results.

A fundamental assumption underlying the design of `HEXPOSE` is that a user's exposure is inversely related to the number of other users whose behavior is indistinguishable from that of user i, that is, whose metric vector representations fall within i's hyperbox. We refer to these users as **neighbors** of i, and denote their count as NC_i. Users with smaller NC_i exhibit more distinctive behavioral patterns relative to others and are, therefore, considered more exposed.

We determine NC_i by first identifying the neighbors of i, one dimension (metric) at a time. That is, for each metric m_r and each candidate neighbor $j \in U$, we define a binary variable $I_{i,j}^r$ to indicate whether j's behavior, with respect to metric m_r, is similar to i's, given the tolerance parameter v as follows:

$$I_{i,j}^{r} = \begin{cases} 1, \text{ if } (1-v) \times m_r^i \leq m_r^j \leq (1+v) \times m_r^i, \ \forall j \in U, \ i \neq j \\ 0, \text{ otherwise} \end{cases} \tag{1}$$

Next, we identify if a user j is a neighbor of i by defining binary variable $B_{i,j}$ to indicate whether j falls within i's hyperbox, that is, whether j is similar to i in *all dimensions*:

$$B_{i,j} = \begin{cases} 1, \text{ if } \sum_{r=1}^{|M|} I_{i,j}^{r} = |M|, \quad \forall j \in U, \ i \neq j \\ 0, \text{ otherwise} \end{cases} \tag{2}$$

Lastly, the number of neighbors of i is defined as:

$$NC_i = \sum_{\substack{j=1 \\ j \neq i}}^{|U|} B_{i,j} \tag{3}$$

We use NC_i to quantify the exposure of user i and leverage it in the definition of i's *exposure score*, $\texttt{HEXPOSE}_i$.

A straightforward approach is to define a *binary exposure score*: A user is considered (i) exposed if the user has no neighbor in their hyperbox ($NC_i = 0$) and thus $\texttt{HEXPOSE}_i = 1$, or (ii) protected if $NC_i > 0$, resulting in $\texttt{HEXPOSE}_i = 0$. This binary classification is easy to interpret and consistent with most prior studies on user exposure in mobility data [11, 23]. However, in practice, a user with only one or a few neighbors may still be at risk, especially if those neighbors' behaviors change over time. As the number of neighbors increases, the user becomes less exposed. Yet, the marginal protective benefit of each additional neighbor is expected to diminish. In other words, *the first few neighbors significantly reduce exposure, while further additions yield progressively smaller gains.* To better capture this phenomenon, we propose a *continuous exposure score* for user i, defined as a decreasing function of NC_i. This formulation allows for a gradual and nuanced characterization of user risks beyond a binary classification. To reflect the diminishing returns effect, the exposure function is designed to exhibit non-linear decay. In this work, we adopt the following inverse decay function, as it is a parameter-free function that gradually decays, better preserving differences among users with low neighbor counts (i.e., high-risk users):

$$\texttt{HEXPOSE}_i = \frac{1}{NC_i + 1} \tag{4}$$

Assessing user exposure as a *continuous value* enables the prioritization of privacy protection interventions for the most exposed users or underlying behavioral patterns, which is particularly valuable when resources are constrained. It also supports flexible categorization of users based on ranges of $\texttt{HEXPOSE}_i$ values.

Lastly, we note that HEXPOSE is a *generic* technique that can be instantiated with different sets of mobility metrics M (cf. §4.2). The framework operates

independently along each behavioral dimension, without combining metrics when defining behavioral neighborhoods. As a result, it is naturally robust to correlations among metrics: each dimension is evaluated against its own tolerance bounds, and two users are considered similar only if all dimensions satisfy the similarity criterion. This prevents any single metric from disproportionately influencing the outcome, even in the presence of correlated or duplicated features.

We further discuss HEXPOSE computational scalability in Appendix §A.

3.2 Interpretability

A key advantage of HEXPOSE lies in its ability to identify what makes a user more/less exposed. Specifically, HEXPOSE offers **interpretability** by: (i) directly quantifying a user's exposure through the occupation of her hyperbox (cf. Equation 3); (ii) explicitly revealing the specific dimensions or mobility metrics contributing to the user's exposure based on the relative tolerance v, and (iii) enabling a straightforward estimation of the effort required to achieve protection in each dimension. This contrasts with aggregation-based approaches, where interpretability may be obscured by aggregate measures, such as the average distance used by MoBES [14]. This section explores the interpretability of HEXPOSE by focusing on exposed users, as they represent the most privacy-critical and analytically informative scenario.

For the sake of simplicity, we first consider the binary exposure setting in which a user i is considered fully *exposed* ($\texttt{HEXPOSE}_i = 1$) when no neighbor is present within i's hyperbox. The user transitions to a *not exposed* state as soon as a single neighbor enters the hyperbox, reducing $\texttt{HEXPOSE}_i$ below 1. This discrete boundary naturally links the interpretation of a user's exposure to her **closest potential neighbor**, i.e., the other user who would be the first to enter i's hyperbox if the threshold v were slightly relaxed. Identifying this potential neighbor reveals not only *how distant user i is to gaining protection, but also which behavioral dimension contributes most to i's exposure.* We formalize this point below and later extend the analysis to the continuous exposure setting.

To assess how distant user i is to protection, we define the *hyperbox-relative distance* between i and another user j as:

$$P_{i,j} = \max_{m_r \in M} \left(\frac{|m_r^i - m_r^j|}{m_r^i} \right) \tag{5}$$

Two important observations must be made regarding Equation 5. First, this is a **relative and asymmetric** distance. It captures, from the perspective of a user i, the deviation from the behavior of her neighbors, thus, $P_{i,j} \neq P_{j,i}$, as $P_{j,i}$ captures the distance from the perspective of j. Second, because it is relative, the difference is divided by the metrics of i, i.e., m_r^i. Thus, to avoid infinite values, we must ensure that these metrics respect the following constraint: $m_r^i > 0$. We do so by adding a small constant (e.g., 1×10^{-6}) to every metric.

User j is considered a neighbor of user i if and only if $P_{i,j} \leq v$, meaning that all relative deviations fall within the allowed threshold v. Thus, $P_{i,j}$ quantifies

how far user j lies from the boundary of i's hyperbox: values greater than v indicate that j falls outside the hyperbox (i.e., is not a neighbor), while values less than or equal to v indicate that j is within it (i.e., is a neighbor).

The *closest potential neighbor* j^* is the user who minimizes this deviation, i.e., $j^* = \arg\min_{j \in U,\ j \neq i} P_{i,j}$. In our binary exposure scenario, j^* is outside i's hyperbox. From this reference point, our interpretability methodology extracts a multi-layered explanation of exposure as follows:

- **Distance-to-protection.** The corresponding value P_{i,j^*} captures the *maximum deviation* between user i and i's closest potential neighbor j^*, computed across all metrics. It quantifies how much j^*'s current behavior exceeds the allowed tolerance v. If the tolerance were increased to match P_{i,j^*}, user i would gain at least one neighbor and no longer be exposed. This scalar provides a compact, interpretable summary of *how distant* i is to gaining protection.
- **Exposed dimensions.** While P_{i,j^*} summarizes proximity to protection, it does not explain *why* the user is exposed. To uncover this, we identify the subset of behavioral dimensions in which user j^* exceeds the tolerance with respect to user i as follows: $E_i = \left\{ m_r \in M \;\middle|\; \frac{|m_r^i - m_r^{j^*}|}{m_r^i} > v \right\}$. E_i groups the dimensions responsible for user i's exposure. Identifying them is essential as the effort required to mitigate exposure may not be uniform across dimensions. Some metrics may already fall within the tolerance and require no adjustment, while others may be only marginally exposed. Without this knowledge, privacy-preserving interventions risk being either overly costly or inefficient.
- **Contribution to exposure.** To measure how strongly each exposed dimension contributes to the exposure, we compute an *exposure deviation score:* $D_i^r = \frac{|m_r^i - m_r^{j^*}|}{m_r^i}$. This value quantifies how far each dimension r is from satisfying the tolerance. Dimensions with D_i^r close to v require only minor adjustments, while higher values reflect more significant behavioral exposures.

Ranking the elements of E_i by their deviation scores D_i^r yields a detailed *exposure profile* for user i, identifying the behavioral dimensions that contribute to exposure and the effort required to mitigate it. Hence, `HEXPOSE` supports **exposure remediability** by linking exposure to specific behavioral factors and quantifying the effort needed for protection, providing actionable insights to reduce exposure. This enables targeted–rather than uniform–protection strategies focused on the relevant dimensions. While promising for tailored protection, their full development is beyond the scope of this paper.

Finally, the interpretability methodology discussed above, though illustrated in a binary exposure setting, naturally generalizes to continuous cases where exposure is defined as a decreasing function of the number of neighbors (e.g., $\texttt{HEXPOSE}_i = \frac{1}{NC_i+1}$). In such settings, a user i becomes exposed when her $\texttt{HEXPOSE}_i$ value crosses the boundaries of the defined threshold, regardless of how many neighbors she already has. We identify the *next potential neighbor*, i.e., the user outside the hyperbox whose inclusion would most effectively reduce exposure (and remove the exposed status), and apply the same interpretability steps from

this reference point. As in the binary case, the continuous setting preserves the methodology's interpretability power while supporting more flexible, score-based definitions of exposure.

4 Experimental Setup

We now detail HEXPOSE practical application by presenting the metrics used in this study to build the behavioral space (§4.1), the datasets on which our analysis is based (§4.2), and the tolerance parameter that governs behavioral similarity and exposure sensitivity in our experiments (§4.3).

4.1 Behavioral metrics

The $|M|$-dimensional behavioral space provides a computationally efficient and expressive alternative to raw trajectories by representing users through compact vectors of complementary mobility metrics, thereby enabling the detection of otherwise hidden behavioral patterns. While we *do not aim* to propose new mobility metrics, we select 13 widely used metrics that capture well-established aspects of human mobility behavior and provide complementary views across spatial, temporal, and structural dimensions. Jointly, such metrics provide a richer and more nuanced understanding of individual mobility patterns, enabling meaningful exposure assessment. These metrics are chosen because they can be computed directly from individual trajectories without requiring external or contextual data, ensuring a balanced and interpretable representation of behavioral variability. For the sake of fair comparison, we consider the same metrics used in [14]. While additional metrics could be incorporated, our goal is not to exhaustively explore the space of possible metrics, but to demonstrate the effectiveness of HEXPOSE on a representative and commonly used set.

Depending on the type of data on which they rely, the selected metrics are grouped into three categories discussed hereafter.

Spatial features rely only on geolocation information:

- *Radius of gyration (RoG)*: captures the characteristic confinement zone of the user mobility, measured as the average distance to the center of mass of the user's trajectory [15].
- *2k Radius of gyration (2kRoG)*: captures the displacement relative to the user's strongest routine, namely home and work [25].
- *Maximum displacement (MDIST)*: maximum distance between two consecutive visits [21].
- *Avg/Stddev of jump lengths (AJL, SJL)*: average and standard deviation of the distances between consecutive visits [15].
- *Number of visits (NOV)*: number of records of the user [24].
- *Number of unique locations (NUL)*: captures the user's mobility diversification in terms of locations visited [15].

Temporal features rely only on visiting times:

- *Avg/Stddev of waiting times (AWT, SWT)*: average and standard deviation of the time spent in each location [29].

Structural features depend on time-ordered visits, relying on both spatial and temporal information:

- *Diversity (DIV)*: captures how diverse a user's visitation pattern is in terms of subsequences of visited locations [9].
- *Regularity (REG)*: captures if a user is prone to return to previously visited location [10].
- *Stationarity (STAT)*: captures whether a user often stays continuously in the same location [10].
- *Entropy (ENT)*: captures the uncertainty in mobility [29].

4.2 Datasets

We evaluate `HEXPOSE` using large-scale, fully anonymized CDR datasets from the cities of Shanghai and Shenzhen, in China, which exhibit complementary characteristics. The Shenzhen dataset records user locations only during communication events, resulting in higher temporal sparsity, whereas the Shanghai dataset logs locations hourly regardless of activity. In addition, the two datasets cover observation areas of different sizes, with Shanghai spanning a substantially larger region. While the Shanghai dataset covers an area of approximately 14,000 km^2, the Shenzhen dataset covers around 997 km^2, as estimated using the convex hull of the observed locations.

Both datasets underwent a common pre-processing pipeline, including hourly time discretization, a 200m $\times$ 200m grid tessellation, removal of users with fewer than 100 records or fewer than 10 active days, and trajectory completion by inferring missing locations around home and work [4, 9]. In the Appendix A, we present the Table 2, which summarizes the resulting datasets.

Our analysis reveals differences in mobility patterns between the two datasets, driven by geographic and data-collection differences, which are consistent with prior observations [3, 12]. Users in the Shanghai dataset, with broader geographic coverage, are generally less spatially constrained, exhibiting higher spatial metrics (RoG, MDIST, AJL, SJL) and more visits to unique locations. However, the Shenzhen dataset shows a large number of users with limited movement, often commuting between home and work. At the same time, a small group exhibits extreme behavior, traveling much more or visiting many more places than the average person. This leads to greater variability in various metrics across the users. Structural metrics exhibit broadly similar trends across both datasets, but with some differences reflecting variations in data collection procedures: Shanghai captures more consistent routines (higher stationarity, fewer outliers), whereas Shenzhen users display higher entropy and diversity among frequent movers, suggesting movement sequences that are harder to predict. These complementary characteristics make the datasets well-suited for evaluating `HEXPOSE` under contrasting mobility and data-quality conditions.

4.3 Tolerance parameter settings

HEXPOSE relies on a single parameter v, which defines the maximum relative deviation allowed per behavioral metric between a user i and any neighbor in the behavioral space. The choice of v is context-dependent and reflects the desired sensitivity to behavioral differences. Smaller values of v enforce stricter exposure criteria, isolating users even with minor behavioral variations, whereas larger values of v allow greater variation and thus a lower notion of exposure. As HEXPOSE is a continuous score (cf. Equation 4), increasing v can only reduce or maintain exposure: for any $v_2 > v_1$, we have $U_{exposed}(v_2) \subseteq U_{exposed}(v_1)$.

In our empirical analysis, we consider $v \in 5\%, 10\%, 20\%$, representing increasing tolerance levels for behavioral similarity: (i) $v = 5\%$ corresponds to high-risk settings requiring strict privacy guarantees, as commonly seen in health data anonymization [19]; (ii) $v = 10\%$ reflects moderate-risk contexts, balancing privacy and utility [5]; (iii) $v = 20\%$ fits low-risk scenarios where greater variation is acceptable [30]. These values illustrate how HEXPOSE can be calibrated to different levels of privacy sensitivity in risk assessments. on

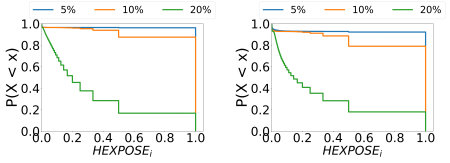
5 Empirical Findings on Behavioral Exposure

We now present an empirical analysis of behavioral exposure as captured by HEXPOSE. Our investigation follows a sequence of analytical steps that progressively characterize exposure, how severe it is, and which behavioral factors drive it in real-world mobility data.

S1- We begin by examining how exposure prevalence and score distributions evolve as the tolerance parameter v varies, highlighting the sensitivity of behavioral exposure to similarity thresholds (§5.1).
S2- We then quantify the severity of exposure by measuring the distance-to-protection, which captures the behavioral change required for a user to no longer be exposed (§5.2).
S3- Next, we analyze the dimensionality of exposure by identifying how many behavioral metrics simultaneously contribute to a user's exposure, and which dimensions are most frequently involved (§5.3).
S4- We further examine how behavioral exposure complements traditional location-based privacy analyses, illustrating which exposure patterns remain invisible to location-centric approaches (§5.4).
S5- Finally, we relate behavioral exposure to a concrete re-identification setting, analyzing its relationship with the success of a simple location-based attack and highlighting the complementary risks captured by HEXPOSE (§5.5).

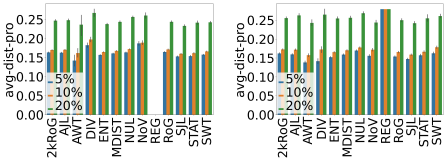
5.1 Effect of tolerance on exposure prevalence

Fig. 2 shows the distribution of individual exposure, i.e., $\texttt{HEXPOSE}_i$ values, for increasing tolerance values v, computed using the inverse decay formulation (Equation 4). At low tolerance levels ($v \leq 10\%$), exposure is widespread: more



(a) Shanghai (b) Shenzhen

Fig. 2: Distribution of HEXPOSE across tolerance values v (5%, 10% or 20%).



(a) Shanghai (b) Shenzhen

Fig. 3: Average distance-to-protection (avg-dist-pro) across primary exposure drivers.

than 80% of users in both datasets have no behavioral neighbors, corresponding to full exposure ($\texttt{HEXPOSE}_i = 1$). Increasing tolerance to $v = 20\%$ substantially reduces exposure, but does not eliminate it. While over 80% of users in both datasets have at least one neighbor ($\texttt{HEXPOSE}_i < 1$), a large fraction remains weakly protected: 45% of Shanghai users and 41% of Shenzhen users have $\texttt{HEXPOSE}_i > 0.2$, corresponding to at most four neighbors.

For a fixed tolerance, users in Shenzhen consistently exhibit lower exposure scores than users in Shanghai. This indicates greater behavioral overlap in Shenzhen, likely due to more homogeneous daily routines for a large fraction of users, such as recurring home–work commuting patterns (cf. §4.2). In contrast, Shanghai users display more diverse mobility behaviors beyond routine locations, resulting in higher distinguishability and increased exposure.

S1 Finding: Behavioral exposure is highly sensitive to the tolerance parameter. Even moderate tolerance values leave a large fraction of users exposed, while exposure values vary substantially across datasets, reflecting differences in behavioral homogeneity.

5.2 Severity of behavioral exposure

We next assess the severity of behavioral exposure using the *distance-to-protection*, which measures how much a user's behavior would need to change to become indistinguishable from others (cf. §3.2). From this analysis and in the remainder, we consider a binary assessment of exposure. Besides, we focus on the primary exposure driver (i.e., the dimension with highest deviation), while all exposed dimensions are examined in Appendix B.

Fig. 3 reports the average distance-to-protection for exposed users across tolerance values and behavioral metrics. For $v = 5\%$ and $v = 10\%$, the distributions are nearly identical across both datasets and metrics. In these settings, most exposed users have distances around 0.15, indicating that their behavioral deviation exceeds the allowed tolerance by a noticeable margin relative to v, even though it remains moderate in absolute terms. At $v = 20\%$, distances diverge more clearly across metrics and datasets, revealing heterogeneity in exposure

severity. In Shanghai, *DIV* (diversity) and *NoV* (number of visits) exhibit larger distances, suggesting that users exposed along these dimensions deviate more substantially from the population norm. In Shenzhen, *DIV* and *NUL* (number of unique locations) play a similar role, reflecting stronger sensitivity to variations in visited locations under event-based sampling. In contrast, *REG* (regularity) shows consistently small distances in both datasets, with nearly all users lying close to their neighbors along this dimension. As a result, REG contributes very little to exposure, even when users are flagged as exposed.

A complementary analysis that considers *all* exposed dimensions (beyond the most severe one) is presented in Appendix B, confirming that the magnitude of exposure varies substantially across metrics and users.

S2 Finding: *Distance-to-protection* shows that most behaviorally exposed users lie relatively close to the exposure boundary, while a subset exhibits large deviations along specific metrics (e.g., diversity and spatial spread), indicating stronger behavioral distinctiveness.

5.3 Multi-dimensional nature of behavioral exposure

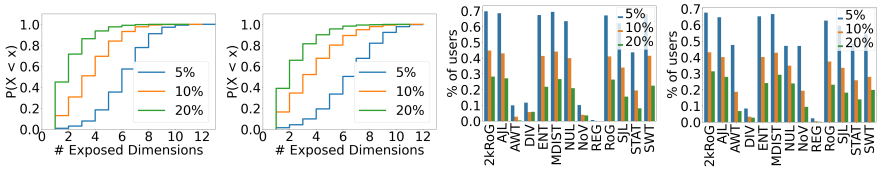
While the distance-to-protection focuses on the most exposed metric, behavioral exposure is often multi-dimensional. To capture this, we analyze the set of exposed dimensions E_i for each user (cf. §3.2). Figures 4a–4b report the distribution of exposed dimensions per user, while Figures 4c–4d show the fraction of users for whom each metric exceeds the tolerance v and belongs to E_i.

Across both datasets and all tolerance values, exposure exhibits substantial heterogeneity. Even under a relaxed threshold ($v = 20\%$), more than half of users exhibit multiple exposed dimensions (up to six), indicating that exposure rarely arises from a single behavioral trait. Most metrics contribute to exposure for a non-negligible fraction of users, with spatial metrics (except NoV) and several structural metrics (notably ENT and STAT) appearing particularly frequently. Moreover, metrics that often appear in E_i also tend to be dominant contributors to exposure, as reflected by their larger distance-to-protection values (Figures 2a and 2b). This highlights that behavioral exposure is inherently multi-faceted and cannot be mitigated by protecting a single dimension alone.

S3 Finding: Behavioral exposure is inherently multi-dimensional. A majority of exposed users violate the tolerance threshold along multiple metrics simultaneously, even under relaxed settings. As a result, mitigating exposure requires accounting for several behavioral dimensions rather than addressing a single dominant feature.

5.4 Complementarity of behavioral and location-based exposures

We examine how `HEXPOSE` complements two existing measures, MoBES [14] and Uniqueness [11], which capture different notions of exposure. While MoBES and `HEXPOSE` operate in a behavioral space, Uniqueness focuses on location-based user



(a) Shanghai (b) Shenzhen (c) Shanghai (d) Shenzhen

Fig. 4: Exposed dimensions: (a),(b) Number of exposed dimensions. (c),(d) Share of users for which each dimension is larger than v (i.e, metric appears in E_i).

exposure. We assess which exposure patterns are uniquely revealed by HEXPOSE in a binary assessment. Table 1 summarizes the overlap between users flagged as exposed by HEXPOSE, MoBES, and Uniqueness across datasets.

HEXPOSE and MoBES. MoBES estimates exposure using the average distance to the k-nearest neighbors in the behavioral space, flagging users whose average distance exceeds a threshold τ. For alignment, we use the same behavioral metrics (cf. §4.1), set $k = 1$, and consider $\tau = v \in \{5\%, 10\%, 20\%\}$. While MoBES aggregates deviations across dimensions, HEXPOSE flags exposure whenever a user exhibits a large deviation along any single dimension.

Across both datasets, HEXPOSE consistently identifies all users flagged by MoBES, while additionally capturing users whose exposure is driven by localized behavioral deviations. These cases remain undetected by MoBES because their average distance remains low despite strong deviations in individual metrics. This highlights a key difference in design: while both approaches rely on behavioral representations, HEXPOSE departs from aggregation-based formulations by introducing a hyperbox-based representation that captures distinctiveness along individual dimensions, enabling it to reveal nuanced exposure patterns that remain hidden under the averaging mechanism of MoBES. Notice that MoBES does not identify any patterns that are not also captured by HEXPOSE, resulting in a constant value of 0 in the *Only MoBES* column (Table 1). Rather than indicating MoBES ineffectiveness, this result suggests that the exposure captured by MoBES is fully contained within that identified by HEXPOSE.

HEXPOSE and Uniqueness. Uniqueness measures exposure based on whether a user's spatio-temporal trajectory within a time window is unique. Following prior work, we use a window size of $t_w = 4$.

Uniqueness flags a large fraction of users as exposed, particularly in Shanghai, where trajectories are more spatially diverse. For smaller tolerances ($v = 5\%, 10\%$), HEXPOSE and Uniqueness identify largely overlapping sets of exposed users, reflecting conservative exposure assessments. As v increases, HEXPOSE becomes more selective, focusing on users with strong behavioral deviations, while Uniqueness remains strictly location-based. Importantly, HEXPOSE characterizes exposure through behavioral dimensions beyond visited locations, revealing how users differ in regularity, diversity, or temporal structure. At the same time, some users

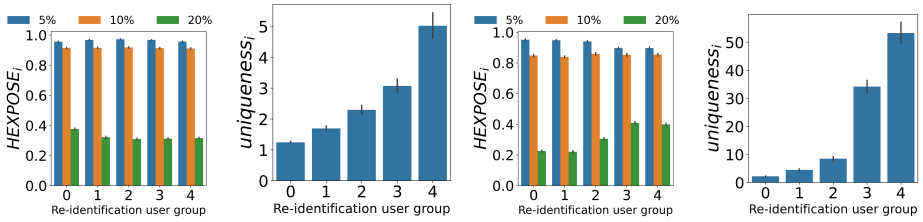
Table 1: Fraction (%) of users flagged as exposed by different techniques (relative to the union on both techniques). MoBES parameters are k=1, $\tau = v$.

(a) MoBES x HEXPOSE

City	$\tau = v$	Only MoBES	Both	Only HEXPOSE
Shanghai	5%	0.000	0.538	0.462
	10%	0.000	0.020	0.980
	20%	0.000	0.001	0.999
Shenzhen	5%	0.000	0.506	0.494
	10%	0.000	0.045	0.955
	20%	0.000	0.001	0.999

(b) Uniqueness x HEXPOSE

City	v	Only Uniqueness	Both	Only HEXPOSE
Shanghai	5%	0.003	0.981	0.015
	10%	0.09	0.899	0.011
	20%	0.825	0.173	0.002
Shenzhen	5%	0.002	0.884	0.114
	10%	0.119	0.794	0.087
	20%	0.786	0.191	0.022



(a) HEXPOSE (b) Uniqueness (c) HEXPOSE (d) Uniqueness

Fig. 5: Relationship between re-identification risk (increasing with user group) and exposure measures: (a)-(b) Shanghai. (c)-(d) Shenzhen.

flagged as unique are not captured by HEXPOSE, highlighting that location-based and behavior-based exposure capture complementary risks.

S4 Finding: Behavioral exposure captured by HEXPOSE complements existing exposure measures. Compared to MoBES, HEXPOSE reveals localized behavioral deviations masked by averaging. Compared to Uniqueness, it characterizes exposure beyond visited locations, highlighting behavioral vulnerabilities not captured by spatio-temporal distinctiveness alone.

5.5 Relationship with re-identification attack

To relate behavioral exposure to a concrete adversarial setting, we evaluate its connection to a simple re-identification attack based on partial location knowledge, following prior work [23,26]. In this setting, re-identification corresponds to linking an observed mobility trace to a user's identity using auxiliary information. For each user, the attacker is assumed to know her most visited location (top-1). All users sharing this most visited location are considered as candidate matches, forming a set of size m. We report the attack success as the attacker's *certainty gain*, defined as $1 - 1/m$, which captures how confidently the attacker can narrow down the identity. Within each dataset, users are then partitioned into five groups (0–4) with increasing re-identification risk, where Group 0 represents the

lowest-risk users and Group 4 the highest-risk ones. The groups were based on the **quantiles** of the *certainty gain* distribution. Once again the corresponding ranges are, from lowest to highest risk: Shanghai [50.00–99.24]%, [99.24–99.62]%, [99.62–99.75]%, [99.75–99.84]%, [99.84–99.95]%; Shenzhen [66.66–99.62]%, [99.62–99.79]%, [99.79–99.89]%, [99.89–99.93]%, [99.93–99.95]%.

Fig. 5 shows the relationship between re-identification risk groups and exposure. As expected, Uniqueness exhibits a strong monotonic increase with attacker certainty, confirming that the attack directly exploits spatio-temporal distinctiveness. In contrast, `HEXPOSE` remains relatively stable across re-identification risk groups for strict tolerance values ($v \leq 10\%$), showing limited alignment with attack success despite high exposure. Only at higher tolerance ($v = 20\%$) does a clearer increasing trend emerge, especially in Shenzhen. This highlights that behavioral exposure captures a complementary notion of risk: users can be behaviorally distinctive without being easily re-identified under location-based matching, revealing vulnerabilities not exploited by current attacks.

S5 Finding: Re-identification risk based on top visited locations strongly correlates with Uniqueness but only weakly with behavioral exposure captured by `HEXPOSE`. This shows that `HEXPOSE` captures complementary privacy risks, revealing behaviorally distinctive users that remain undetected by location-based reidentification attacks.

6 Conclusion and Discussion

This paper introduced `HEXPOSE`, a novel framework for quantifying behavioral exposure in mobility data. By representing users in a multidimensional behavioral space and defining hyperbox-based similarity regions, `HEXPOSE` provides a fine-grained and interpretable assessment of exposure, overcoming the limitations of aggregation-based approaches. Our evaluation on two large-scale CDR datasets shows that `HEXPOSE` identifies all users flagged by MoBES and most of those flagged by Uniqueness, while uncovering additional exposed users missed by both, highlighting its ability to reveal complementary exposure patterns.

Practical implications. In practice, `HEXPOSE` is designed as a diagnostic tool for data providers and data curators who need to assess and mitigate privacy risks before data release or analysis. By identifying which users are exposed and, crucially, why, i.e., along which behavioral dimensions, `HEXPOSE` enables targeted protection strategies that go beyond uniform anonymization. For instance, users exhibiting strong behavioral distinctiveness (e.g., in mobility diversity or temporal regularity) could be selectively protected through tailored perturbation, aggregation, or sampling mechanisms, while less exposed users may require lighter protection, thereby preserving overall data utility. Such a practical diagnostic tool also promotes sustainability by employing complex protection mechanisms only when required. More broadly, our results show that behavioral exposure captures vulnerabilities not exploited by current location-based attacks but potentially relevant for more advanced adversaries. As such, `HEXPOSE` can support risk auditing and the design of behavior-aware privacy mechanisms.

A Computational Analysis

One of the key advantages of HEXPOSE is its scalability. When we consider a naive implementation of HEXPOSE, following the equations defined in §3.1, HEXPOSE presents an overall cost of $O(|U|^2 \times |M|)$, since, for each user $i \in U$, checking the number of neighbors within i's hyperbox requires $O(|U| \times |M|)$ operations. Nonetheless, it is important to note that this cost can be significantly reduced by using techniques that reduce the number of comparisons between users.

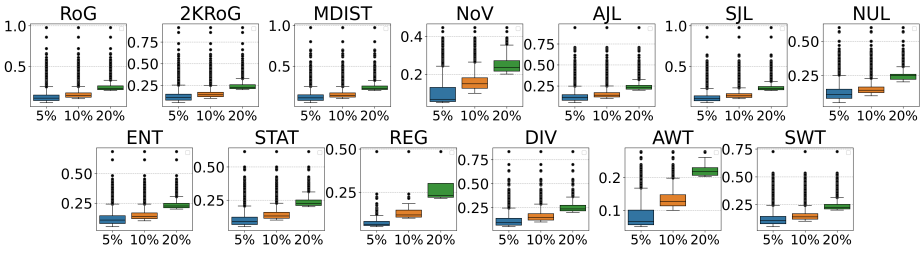
One example would be to avoid comparing a user against all its candidate neighbors by iteratively selecting only users that respect the boundaries of the hyperbox defined by all previously computed metrics. Nevertheless, it is important to note that, in the worst case, such a technique would have a complexity similar to that of a naive implementation.

Another possibility is implementing more sophisticated data structures, such as K-D trees and R-Trees [27], which are used to optimize k-nearest neighbors-based techniques. These techniques work by partitioning the space into hyperplanes defined by the data's dimensionality (e.g., metrics). Such hyperplanes trim down the search space, reducing the number of comparisons made. These techniques have an overall complexity of $O(|U| \times \log |U|)$, which would consequently reduce the computational complexity of HEXPOSE to the same order. Lastly, in high-dimensional cases ($|M| > 20$), it is recommended to use an approximate nearest-neighbor search [6], as K-D Trees and R-Trees perform poorly, effectively reducing them to a near-brute-force search.

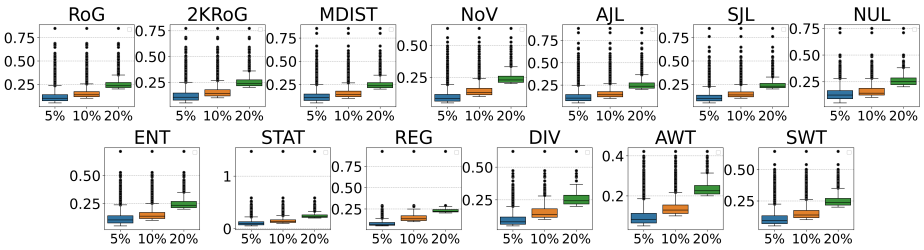
Metric Space: It is important to note, however, that the computation of HEXPOSE is not restricted to the search within the metric space, but also depends on how this metric space is constructed. Assuming that each user's records are temporally ordered, all metrics, except diversity, exhibit linear complexity with respect to the trajectory length. Specifically, the computational complexity per user is $O(|L| \times |U|)$, where $|L|$ denotes the length of the user's trajectory.

On the other hand, diversity incurs a higher computational cost than the other metrics considered. Diversity explores combinations of locations visited by each user, which is computed in $O(L^2)$, and can be efficiently computed in $O(|L|log|L|)$ using a suffix array combined with the longest common prefix (LCP) array [31]. As such, the diversity metric dominates the computation overhead of the metrics. Consequently, the overall computational cost of HEXPOSE using the space of metrics presented in §4.1 is $O((|L| \times |U|) + O(|L|log|L|) \times |M|)$.

It is important to highlight that the metric space can be custom-designed according to the preferences of the data owner. Thus, in scenarios where large volumes of data are available, such as for mobile carriers, it is possible to reduce the overall cost using different strategies, such as (i) reducing the size of the metric space, (ii) selecting only metrics with linear complexity, (iii) parallelizing the computation of the metrics and the user comparisons, and (iv) sampling user trajectories to capture behavior using fewer records.



(a) Shanghai



(b) Shenzhen

Fig. 6: Distance-to-protection in all exposed dimensions, i.e., largest distance to the closest neighbor. Note that the metrics' y-axes are in different scales.

B Exposure drivers across behavioral metrics

Figure 6 reports the distance-to-protection for all exposed dimensions, i.e., metrics whose values exceed the tolerance v, across all tolerance levels and both datasets. Metrics that frequently act as primary exposure drivers (e.g., 2kRoG) exhibit substantially larger deviations from v, indicating that exposure along these dimensions is harder to mitigate. In contrast, metrics that rarely dominate exposure, such as STAT and REG (Figures 4c, 4d), show average deviations close to v, confirming their limited contribution to overall exposure.

This analysis also clarifies dataset-specific exposure patterns. In Shanghai, spatial metrics such as RoG, MDIST, and SJL exhibit deviations close to 1, indicating broader movement ranges and greater behavioral diversity. In Shenzhen, greater variation is observed in metrics such as STAT, REG, and NoV, highlighting the impact of the event-based data collection process. Users with fewer communication events are observed at more dispersed locations, leading to lower stationarity and regularity and, consequently, higher exposure.

Overall, metric contributions to exposure vary substantially across datasets, reflecting differences in mobility behavior and data collection processes. While some metrics play a limited role on average, they can become highly discriminative in specific contexts. For example, regularity and diversity may sharply distinguish residents from visitors in touristic settings, while the relevance of AWT differs markedly between Shanghai and Shenzhen.

C Dataset Statistics

Table 2: Summary statistics of the Shanghai and Shenzhen mobility datasets, describing and highlighting differences in dataset scale, spatial diversity, and user-level activity (95% CIs have half widths < 0.01).

Metric	**Shanghai**	**Shenzhen**
Total # users	58502	31859
Total # days	10	15
Total # unique locations	6064	8669
Avg. # unique locations per user	12.4	9.67
Avg. # unique locations per user per day	3.37	2.8
Avg. # records/visits per user	180.37	136.01
Avg. # records/visits per user per day	18.03	9.25

Acknowledgments

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